Project/ASG DSA SEC 03

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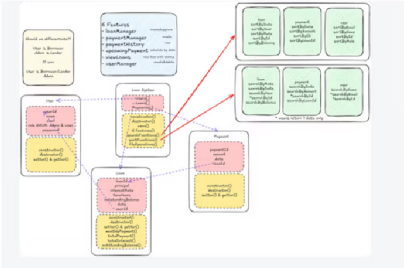
3

GROUP 2

ww.to

comment: Good job!

Title : Loan & Installment Payment System



Proposed function:
1.loanCreator
2.paymentCreator
3.paymentHistory
4.upcomingPaymen
5.viewLoans
6.userManager

Searching
- Linear Search
  - loan: date[], userId[]
  - user: name[]
  - payment: paymentId[]
- Binary Search
  - loan: loanId
  - user: userId
  - paymentL amount, userId, loanId, date

Sorting
- Quick Sort
  - loan: loanId, date, amount, userId
  - user: userId, name
  - payment: amount, userId, loanId, date

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Project/ASG DSA SEC 03

1

2

3

Course Registration & Timetable System

Student function

1. Course Registration
2. View all courses
3. Search for courses (search by different variable using sequential searching)
4. View Personal Timetable (sort by different variable using bubble sort, insertion sort and selection sort)

Admin function

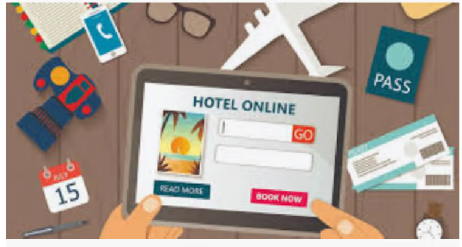
1. Admin login authentication
2. Manage Course Timetable
3. Display All Courses (sort by different variable using bubble sort, insertion sort and selection sort)
4. Add or Delete Courses

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SAMPLE MINIMUM 2 FUNCTIONS EACH student

TITLE: HOTEL BOOKING

PROPOSE FUNCTIONS
1. MAKE BOOKING
2. FUNCTION 2
3. FUNCTION 3
4. FUNCTION 4
5. FUNCTION 5



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SAMPLE MINIMUM 2 FUNCTIONS EACH STUDENT



Faculty of Computing
UNIVERSITI TEKNOLOGI MALAYSIA

Programming Technique II
(SECJ1013)

SEMESTER x 2023/2024-2

Project and Assignment Software Documentation:
UTM Hall Management System

by
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Fatema Junaid (A22CS0058)

SECTION 06

Lecturer:
Ts. Dr. Johanna Binti Ahmad

Date: 30 June 2024

PDF

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comment: Good Job!

GROUP 1

Title: Clinic Queue Management System

Functions Proposed:
1. Add Patient to Queue
2. Remove Patient
3. Sort Patient
4. Search Patient
5. Display Patient Info

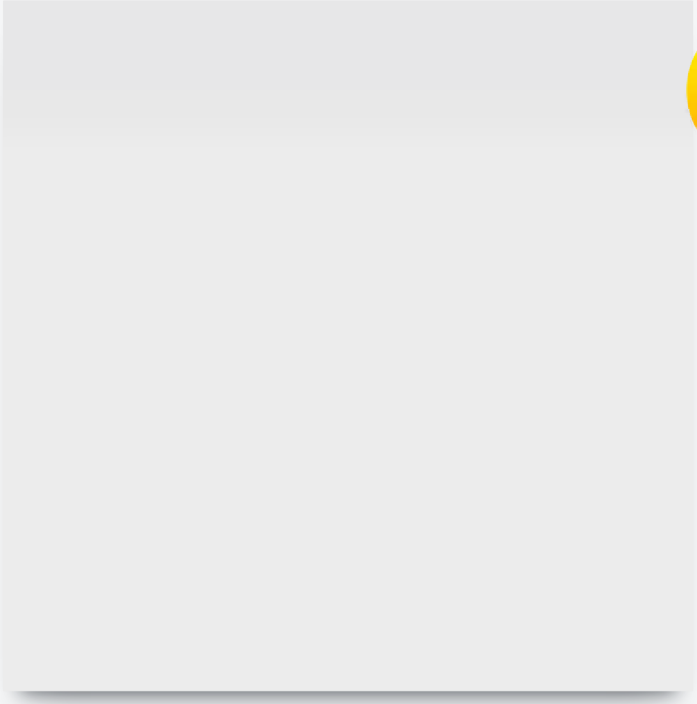
Sorting
1) Bubble sort - sort by priorityDesc
2) Bubble sort - sort by priorityAsc
3) Selection sort - sort by Name
4) Insertion sort - sort by patientID

Searching
1) Binary search - search by ID
2) Linear search - search by ID

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📜 GROUP 2

comment: Good Job!



😌

Sorting :
Name - Selection sort
Date - Bubble Sort
Parcel ID - Insertion sort
Status - Bubble sort

Searching :
Parcel ID - Linear search
Customer ID - Binary search

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📜 GROUP 3

TITLE: Course Registration & Timetable System

PROPOSE FUNCTIONS
1. Register Course
2. Sort Courses (Selection)
3. Search Course By Code (Linear)
4. Delete Course
5. Add Course
6. View All Courses

Sorting
1. Selection sort - sort course by code
2. Quick sort - sort course by name
3. Bubble sort - sort course by time

Searching
1. Linear - search course by name
2. binary - search course by lecturer name
3. linear - search course by code
4.

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📜 GROUP 4

Clinic Prescription & Medication System

Function Proposed :

1. login
2. sortInventorybyStock
3. findMedicinebyName
4. doctorPrescribe
5. clerkPrintReceipt
6. clerkRecordVisit
7. clerkGenerateReceipt
8. findMedicinebyId
9. displayAvailableMedicines

Sorting algorithm
Used in function:sortInventorybyStock
Algorithm used: Insertion Sort

Searching Algorithm
Used in function:findMedicinebyName
and findMedicinebyId
Algorithm used: Linear Search

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📜 GROUP 5

comment: Good Job!



Title: Cinema Ticket Reservation system

Proposed functions:
1) loadBookingFromFile
2) addBooking
3) removeBooking
4) searchBooking
5) sortBooking
6) listAllBooking

Sorting
- QuickSort
Booking:
1. Sort by ID
2. Sort by Date
3. Sort by Cinema
4. Sort by Hall

Search
- Linear search
  1. Booking by CinemaID

- Binary search
  1. Search booking by BookingID

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📜 GROUP 6

Title: Vehicle Parking Management system

Proposed functions:

1) loadFromFile
2) addVehicle
3) removeVehicle
4) searchVehicleByPlate
5) sortVehiclesByPlate
6) listAllVehicles

Roles Involved:

1. Admin:
   1. add vehicle
   2. remove vehicle
   3. View vehicle
   4. Sort Vehicle
   5. Search Vehicle
   6. Load data from file
2. User:
   1. View all vehicle
   2. Search for vehicle
   3. Check slot availability

Sorting:

1. Bubble Sort
   - Sort by parking slot
   - Sort by time enter
2. Quick Sort
   1. Sort by plate number

Search:

1. Linear Search(search by plate number)
   - Search by parking slot number
   - Search by type of vehicle
   - Search by plate number

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GROUP 7

Inventory Management System
1) ViewALLInventoryList
2)SortListByName
3)SortListByPrice
4)SortListByQuantity
5)SearchStockByName
6)UpdateName
7)UpdatePrice
8)UpdateQuantity

Sort: Using Bubble Sort (all asc)
Search: Using Linear Search

User Role involved:
1)User: comment: Good Job!
- ViewAllInverntory
- SortListByName
- SortListByStock
- SortListByQuantity
- SearchStockByName
- UpdateQuantity

2)Admin:
- ViewAllInventory
- SortListByName
- SortListByStock
- SortListByQuantity
- SearchStockByName
- UpdateQuantity
- UpdateStock
- UpdateName

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GROUP 8

comment: Good Job!

Smart Attendance Tracking System

Sorting

quick sort:
Lecturer : LectID,
Student : StudID,

selection sort :
Course : name

bubble sort :
Lecturer : name
Student : name

insertion sort :

Searching :
Linear search
Student : StudID[],name[]
Lecturer : LectID[],name[]

Binary search
Student : StudID
Lecturer : LectID
Course : name

FUNCTION PROPOSED :
1. sort Attendance
2. Fill attendance
3. Attendance History
4. Attendance status
5. Search Attendance record)
6. Login/Logout Authentication

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📜 Exercise Topic 3

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📜 Exercise Topic 3

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